

Tables of the master thesis

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Table of Abbreviations

Symbol	Meaning	Unit
$\hat{c}$	Geometric centroid proxy	m
c_{obb}	Centre of the object's oriented bounding box	m
c_{pcd}	Centroid of the filtered point cloud	m
d_{min}	Inner bound of the centroid-based grip zone	m
d_{max}	Outer bound of the centroid-based grip zone	m
$\mathbf{f}$	UR10e nominal frontal direction vector	dimensionless
p	Projection of the gripper closing axis onto $\mathbf{f}$	dimensionless
th_{axis_proj}	Axis-alignment projection threshold	dimensionless
$th_{frontal}$	Frontal projection threshold, Filter B1	dimensionless
y_g	Gripper closing axis (y -axis of the gripper frame)	dimensionless
z_g	Gripper approach axis (z -axis of the gripper frame)	dimensionless
z_w	World z -axis	dimensionless
θ_{max}	Maximum angular deviation for the vertical approach constraint, Filter B2	°

Chapter 1

Introduction

Chapter 2

Fundamentals

Table 2.1: Denavit–Hartenberg kinematic parameters for the UR10e, as published by the manufacturer.

Joint i	a_i [m]	d_i [m]	α_i [rad]
1	0	0.1807	$\pi/2$
2	-0.6127	0	0
3	-0.57155	0	0
4	0	0.17415	$\pi/2$
5	0	0.11985	$-\pi/2$
6	0	0.11655	0

Chapter 3

Related Work

Chapter 4

Own Appraoch

Chapter 5

Experimental Setup and Results

Table 5.1: Overall success rates per condition (144 trials each).

Condition	Algorithm	Fingertip	Successes	Rate
Baseline 1	vMF-Contact (unconstrained)	Robotiq original	20 / 144	13.9 %
Baseline 2	vMF-Contact + CoG constraints	Robotiq original	58 / 144	40.3 %
Final	vMF-Contact + CoG constraints	Custom silicone	111 / 144	77.1 %

Table 5.2: Failure mode counts per condition (144 trials each).

Condition	F1 (grip)	F2 (no pose)	F3 (kinematic)	F4 (orient.)	Success
Baseline 1	103	0	11	10	20
Baseline 2	71	1	14	0	58
Final	1	7	25	0	111

Table 5.3: Baseline 1 orientation cluster classification (O1–O8).

Orientation	F1	F2	F3	F4
O1	10	0	0	2
O2	14	0	0	2
O3	18	0	0	0
O4	14	0	0	0
O5	15	0	0	2
O6	13	0	0	1
O7	5	0	11	0
O8	14	0	0	3

Table 5.4: Baseline 2 orientation cluster classification (O1–O8).

Orientation	F1	F2	F3	F4
O1	9	0	0	0
O2	16	0	0	0
O3	5	1	12	0
O4	2	0	0	0
O5	10	0	1	0
O6	11	0	0	0
O7	13	0	1	0
O8	5	0	0	0

Table 5.5: Final orientation cluster classification (O1–O8).

Orientation	F1	F2	F3	F4
O1	1	0	0	0
O2	0	0	0	0
O3	0	3	15	0
O4	0	2	1	0
O5	0	1	0	0
O6	0	0	0	0
O7	0	0	9	0
O8	0	1	0	0

Table 5.6: Baseline 1 position cluster classification (P1–P6).

Position	F1	F2	F3	F4
P1	16	0	2	3
P2	17	0	3	3
P3	16	0	3	1
P4	15	0	3	0
P5	20	0	0	2
P6	19	0	0	1

Table 5.7: Baseline 2 position cluster classification (P1–P6).

Position	F1	F2	F3	F4
P1	11	0	2	0
P2	14	1	1	0
P3	12	0	3	0
P4	11	0	1	0
P5	13	0	3	0
P6	10	0	4	0

Table 5.8: Final position cluster classification (P1–P6).

Position	F1	F2	F3	F4
P1	0	0	5	0
P2	0	3	0	0
P3	0	0	5	0
P4	0	0	4	0
P5	1	1	6	0
P6	0	3	5	0

Appendix

Table A.9: Reachability filter evaluation: per-orientation failure counts and success (8 orientations $\times$ 1 position $\times$ 3 trials = 24 trials total).

Orient.	F1(Grip)	F2(No pose)	F3(Kin.)	PS(Inverted orient.)	Success
O1	0	0	0	2	3
O2	0	0	0	2	3
O3	0	0	0	3	3
O4	0	0	0	0	3
O5	0	0	0	3	3
O6	0	0	0	0	3
O7	0	0	0	3	3
O8	0	0	0	0	3
Total	0	0	0	13	24 / 24